

A counterexample to conjecture 00000007767

Squaring and cubing have infinitely many common periodic points

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Abstract

The complex polynomial maps $f(z) = z^2$ and $g(z) = z^3$ commute and have infinitely many common periodic points on the complex projective line, but have no common positive compositional iterate. They therefore refute the explicit leading equivalence of conjecture 00000007767 in its conventional complex-dynamical interpretation. The proof uses primitive roots of unity of orders 5, 25, 125, $\dots$. The source's unspecified field and iterate-index conventions are disclosed below; the undefined final uniqueness terminology is not needed for this disproof.

1 Statement and conventions

Both the English and Chinese source versions concern commuting polynomials of degree at least two acting on $\mathbf{P}^1$. Their first assertion is

$$|\operatorname{Per}(f) \cap \operatorname{Per}(g)| = \infty \iff f^{\circ m} = g^{\circ n} \text{ for some positive integers } m, n.$$

They then add a common-iterating-map assertion and a uniqueness assertion about a “symmetrized monomial part” in a Ritt decomposition. An exact copy of the bilingual source is bundled as `conjecture.md`.

The source does not specify a base field. This submission explicitly uses complex polynomials acting on $\mathbf{P}^1(\mathbb{C}) = \mathbb{C} \cup \{\infty\}$, the usual classical complex-dynamical setting. This is a disclosed interpretation, not an assumption stated verbatim in the source. No conclusion about every possible arithmetic interpretation is claimed. Positive indices are the standard nonvacuous meaning of “common iterate”: allowing $m = n = 0$ would give the identity for every pair. The zeroth iterate is used only to define iteration recursively. The periodic set means

$$\operatorname{Per}(F) = \{P : \text{there exists an integer } r \geq 1 \text{ with } F^{\circ r}(P) = P\}.$$

The two return times for a common periodic point may differ. They may also depend on the point. Periodicity is not replaced by eventual periodicity. The notation $F^{\circ m} = G^{\circ n}$ means exact equality of self-maps, not equality up to conjugacy.

We refute the forward implication of the displayed equivalence. The full source assertion entails that implication, regardless of how its later uniqueness condition might be made precise. We neither supply a definition of the undefined final terminology nor claim a replacement Ritt theorem.

2 The two actual polynomial maps

Let $f = X^2$ and $g = X^3$ in $\mathbb{C}[X]$. Their projective extensions are

$$F([Z : W]) = [Z^2 : W^2], \quad G([Z : W]) = [Z^3 : W^3].$$

These expressions define maps on projective points: scaling (Z, W) scales both coordinates by the same nonzero power, and the coordinates cannot simultaneously vanish. In the affine chart $[z : 1]$ the maps are $z \mapsto z^2$ and $z \mapsto z^3$, respectively; $\infty = [1 : 0]$ is fixed by both. Their polynomial degrees are two and three. At an affine point both $F \circ G$ and $G \circ F$ act by $z \mapsto z^6$, and both fix infinity. Thus the stated degree and commutation hypotheses hold.

For $d \geq 1$, let $H_d(z) = z^d$ on the affine chart and $H_d(\infty) = \infty$. Induction on $r \geq 0$ gives

$$H_d^{\circ r}(z) = z^{d^r}. \quad (1)$$

The induction step is $(z^{d^r})^d = z^{d^{r+1}}$; the case $r = 0$ uses $d^0 = 1$. This is compositional iteration of the actual maps.

3 Infinitely many distinct common periodic points

For each $k \in \mathbb{N}$, put

$$N_k = 5^{k+1}, \quad \zeta_k = \exp(2\pi i/N_k), \quad P_k = [\zeta_k : 1].$$

The complex number ζ_k has exact multiplicative order N_k . Both 2 and 3 are coprime to N_k . Euler's theorem gives, for $t_k = \varphi(N_k) > 0$,

$$2^{t_k} \equiv 1 \pmod{N_k}, \quad 3^{t_k} \equiv 1 \pmod{N_k}.$$

Whenever $\zeta^N = 1$ and $a \equiv 1 \pmod{N}$, exponent reduction gives $\zeta^a = \zeta$. Applying this observation and (1) yields

$$F^{\circ t_k}(P_k) = P_k, \quad G^{\circ t_k}(P_k) = P_k.$$

Every P_k is therefore a common periodic point with a positive return time.

If $P_k = P_\ell$, equality in the affine chart implies $\zeta_k = \zeta_\ell$. An element has only one exact multiplicative order, so $5^{k+1} = 5^{\ell+1}$. Since powers of 5 are injective in their natural exponent, $k = \ell$. Hence $k \mapsto P_k$ is an injection from $\mathbb{N}$ into $\text{Per}(F) \cap \text{Per}(G)$, proving that this one fixed pair of maps has infinitely many distinct common periodic points. The single point at infinity is not used to supply infinitude.

4 No common positive iterate

Suppose $m, n \geq 1$ and $F^{\circ m} = G^{\circ n}$. Evaluating at the affine point $[2 : 1]$ and using (1) gives

$$2^{2^m} = 2^{3^n}.$$

These are natural numbers viewed in $\mathbb{C}$. Injectivity of the natural-number embedding into $\mathbb{C}$ transfers the equality back to $\mathbb{N}$. Powers of 2 are injective in their exponent, so $2^m = 3^n$. The left side is even because $m \geq 1$, while the right side is odd. This is a contradiction. This argument excludes every pair of positive indices, not merely a bounded search or the case $m = n$.

Theorem 1. *There exist complex polynomials of degree at least two whose projective extensions commute and have infinitely many common periodic points, but whose positive compositional iterates are never equal. Consequently the leading equivalence of conjecture 00000007767 is false in the explicitly stated complex interpretation.*

Proof. Take $f = X^2$ and $g = X^3$. The preceding sections verify all three hypotheses, the infinitude conclusion, and the failure of every common positive iterate. $\square$

5 Formalization and reproducibility

The accompanying Lean 4 project formalizes the complex polynomials and their extensions to the projective line represented by `Option Complex`: `some z` denotes $[z : 1]$ and `none` denotes $[1 : 0]$. Every projective point has exactly one of these forms. The extension of polynomial evaluation is `Option.map`; thus the model includes the point at infinity, actual polynomial evaluation, and compositional iteration.

The namespace is `Conjecture7767`. The main proof correspondence is:

Mathematical step	Lean declaration
Actual polynomial degrees	<code>powerPolynomial_degree</code>
Composition formula	<code>powerMap_iterate_affine</code>
Commuting projective maps	<code>powerMap_commute</code>
Positive root-of-unity return time	<code>primitiveRoot_periodic</code>
Distinct roots	<code>root_injective</code>
Infinite intersection	<code>common_periodic_infinite</code>
No common positive iterate	<code>no_positive_common_iterate</code>
All counterexample requirements	<code>counterexample</code>
Negation of the leading assertion	<code>conjecture_false</code>

The project pins Lean 4.19.0 and Mathlib v4.19.0. A fresh build, direct source replay, and the complete `Check.lean` audit passed with warnings treated as errors. All 18 authored theorems use only `propext`, `Classical.choice`, and `Quot.sound`. The 11 definitions and abbreviations were printed and inspected. An exhaustive inspection of all 40 declarations originating in the compiled proof module also passed, including its 11 generated declarations. There are no custom axioms, admitted proofs, partial declarations, or authored unsafe definitions. Four compiler-generated runtime wrappers are explicitly inventoried separately from the safe logical declarations; none introduces an additional axiom. Compiler and all nine dependency revisions were checked before and after the build. The complete logs and hashes accompany the submission.

The auxiliary checker uses exact integer coefficients and rational angles. It independently compares sparse polynomial iteration with direct integer iteration, tests all 144 pairs $1 \leq m, n \leq 12$, and computes every cycle under doubling and tripling modulo 5, 25, 125, 625. All 780 root records and their two complete cycle decompositions are included. For these four orders the primitive-root counts and periods under either map are respectively 4, 20, 100, 500. Together with the root 1, affine zero, and infinity, the largest tested projective subset has 627 distinct common periodic points. Normal and optimized Python runs both passed 12,875 explicit checks and produced byte-identical output. These finite checks establish neither infinitude nor unbounded nonexistence of common iterates; those statements are proved in Lean without relying on the Python output.

After installing the pinned Lean toolchain, run the following from the submission's `lean` directory (Mathlib cache retrieval requires network access):

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Conjecture7767.lean
lake env lean -DwarningAsError=true Check.lean
```

From the `auxiliary` directory, run `python3 -B run_checks.py`. The full reproduction instructions include the compiled-environment audit and PDF export. Local validation and the bundled independent semantic review are distinct from acceptance by a competition maintainer.